Practicum 9

```
class BankAccount {

    int accountNumber;
    String accountHolderName;
    double accountBalance;

    BankAccount(int accountNumber, String accountHolderName, double
accountBalance) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.accountBalance = accountBalance;
    }

    synchronized void deposit(String userName, double amount) {
        if (amount <= 0) {
            System.out.println("Invalid Deposit Amount");
            return;
        }

        accountBalance = accountBalance + amount;

        System.out.println("User Name: " + userName);
        System.out.println("Deposited Amount: ₹" + amount);
        System.out.println("Updated Balance: ₹" + accountBalance);
    }

    synchronized void withdraw(String userName, double amount) {
```

```
    if (amount <= 0) {
        System.out.println("Invalid Withdrawal Amount");
        return;
    }

    if (amount <= accountBalance) {
        accountBalance = accountBalance - amount;

        System.out.println("User Name: " + userName);
        System.out.println("Withdrawn Amount: ₹" + amount);
        System.out.println("Updated Balance: ₹" + accountBalance);
    } else {
        System.out.println("Insufficient Balance");
    }
}

void displayBalance() {
    System.out.println("Final Account Balance: ₹" + accountBalance);
}

public static void main(String[] args) {

    BankAccount account = new BankAccount(
        1001,
        "Rahul",
        10000
    );

    UserTransaction user1 = new UserTransaction(
```

```
    "User1", account, "Deposit", 2000
);

UserTransaction user2 = new UserTransaction(
    "User2", account, "Deposit", 3000
);

UserTransaction user3 = new UserTransaction(
    "User3", account, "Deposit", 1000
);

UserTransaction user4 = new UserTransaction(
    "User4", account, "Withdrawal", 4000
);

UserTransaction user5 = new UserTransaction(
    "User5", account, "Withdrawal", 3000
);

user1.start();
user2.start();
user3.start();
user4.start();
user5.start();

try {
    user1.join();
    user2.join();
    user3.join();
```

```
            user4.join();
            user5.join();
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        account.displayBalance();
    }
}

class UserTransaction extends Thread {

    String userName;
    BankAccount account;
    String transactionType;
    double transactionAmount;

    UserTransaction(String userName, BankAccount account,
                    String transactionType, double transactionAmount) {

        this.userName = userName;
        this.account = account;
        this.transactionType = transactionType;
        this.transactionAmount = transactionAmount;
    }

    public void run() {

        if (transactionType.equalsIgnoreCase("Deposit")) {
```

```
            account.deposit(userName, transactionAmount);
        }
        else if (transactionType.equalsIgnoreCase("Withdrawal")) {
            account.withdraw(userName, transactionAmount);
        }
    }
}
```

Practicum 10

```
import java.util.ArrayList;
import java.util.Scanner;
```

```
class Student {
    int studentId;
    String studentName;
    String department;
    int semester;

    Student(int studentId, String studentName,
            String department, int semester) {
        this.studentId = studentId;
        this.studentName = studentName;
        this.department = department;
        this.semester = semester;
    }

    public void display() {
        System.out.println("Student ID: " + studentId);
        System.out.println("Student Name: " + studentName);
        System.out.println("Department: " + department);
        System.out.println("Semester: " + semester);
        System.out.println("-------------------------");
    }
}

public class Main {
    static ArrayList<Student> students = new ArrayList<>();

    static void addStudent(Scanner sc) {
        System.out.print("Enter Student ID: ");
```

```
if (!sc.hasNextInt()) {
    System.out.println("Invalid Student ID.");
    sc.nextLine();
    return;
}

int id = sc.nextInt();
sc.nextLine();

for (Student s : students) {
    if (s.studentId == id) {
        System.out.println("Student ID already exists.");
        return;
    }
}

System.out.print("Enter Student Name: ");
String name = sc.nextLine();

System.out.print("Enter Department: ");
String department = sc.nextLine();

System.out.print("Enter Semester: ");

if (!sc.hasNextInt()) {
    System.out.println("Invalid Semester.");
    sc.nextLine();
    return;
```

```
    }

    int semester = sc.nextInt();
    sc.nextLine();

    Student student = new Student(id, name, department, semester);
    students.add(student);

    System.out.println("Student record added successfully.");
}

static void removeStudent(Scanner sc) {
    System.out.print("Enter Student ID to remove: ");

    if (!sc.hasNextInt()) {
        System.out.println("Invalid Student ID.");
        sc.nextLine();
        return;
    }

    int id = sc.nextInt();
    sc.nextLine();

    for (int i = 0; i < students.size(); i++) {
        if (students.get(i).studentId == id) {
            students.remove(i);
            System.out.println("Student record removed successfully.");
            return;
        }
```

```
        }

        System.out.println("Student record not found.");
    }

    static void searchStudent(Scanner sc) {
        System.out.print("Enter Student ID or Student Name: ");
        String input = sc.nextLine();
        boolean found = false;

        for (Student s : students) {
            if (String.valueOf(s.studentId).equals(input)
                    || s.studentName.equalsIgnoreCase(input)) {
                s.display();
                found = true;
            }
        }

        if (!found) {
            System.out.println("Student record not found.");
        }
    }

    static void displayAllStudents() {
        if (students.isEmpty()) {
            System.out.println("No student records available.");
            return;
        }
```

```
    for (Student s : students) {
        s.display();
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int choice = 0;

    do {
        System.out.println("\nStudent Record Management System");
        System.out.println("1. Add Student");
        System.out.println("2. Remove Student");
        System.out.println("3. Search Student");
        System.out.println("4. Display All Students");
        System.out.println("5. Exit");
        System.out.print("Enter your choice: ");

        if (!sc.hasNextInt()) {
            System.out.println("Invalid choice. Please try again.");
            sc.nextLine();
            continue;
        }

        choice = sc.nextInt();
        sc.nextLine();

        switch (choice) {
            case 1:
```

```
                addStudent(sc);
                break;

            case 2:
                removeStudent(sc);
                break;

            case 3:
                searchStudent(sc);
                break;

            case 4:
                displayAllStudents();
                break;

            case 5:
                System.out.println(
                    "Exiting Student Record Management System.");
                break;

            default:
                System.out.println(
                    "Invalid choice. Please try again.");
            }
        } while (choice != 5);

        sc.close();
    }
}
```